SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – ASSIGNMENT

Course Code:	CSA0901	**Course Title:**	**Programming in Java**
CO Assessed:	CO2 – Precision (BL3,BL4)	**Assessment:**	ASSESSMENTTOOL2 - ASSIGNMENT
Weightage:		**Total Marks:**	100
CO1 Statement:	ImplementCollectionFrameworkinterfacessuchasList,ArrayList,Set,Map, HashTable,andHashMapusingiteratorsandgenericstostore,retrieve,and process groups of objects in web applications		
Student Name:	S.Aarathana	**Reg. No.:**	1921260021
Section Batch:	SLOT B	**Date:**	07/08/2026

Code of Conduct

I S.Aarathana (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______________________

Instructions

- This test contains 1 questions Total: 100 Marks.
- No negative marking. Un answered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 30 minutes.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Application of Concepts	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Analytical & Decision-Making Skills	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Solution Design & Approach	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Clarity, Justification & Presentation	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

STUDENT MANAGEMENT SYSTEM

Java Application – String Handling & Java Collection Framework

1. INPUT / OUTPUT CONCEPT

Input: The program uses ***Scanner*** to accept Roll Number, Name, Department, Course, CGPA, menu choice, and substring for searching.

Output: The program displays processed student details, search results, unique departments, Roll Number–CGPA mapping, and course details using ***System.out.println()***.

Input	Example
Roll number	101
Name	aarathana
Department	IT
Course	Java
CGPA	8.7
Substring	priya

2. CONCEPTS USED

Concept	Use in Program
String Handling	trim(), toLowerCase(), split(), substring(), contains() and Character.toUpperCase().
ArrayList	Stores student objects and supports add, display, update and delete.
HashSet	Stores department names without duplicates.
HashMap	Stores Roll Number → CGPA mapping.
Hashtable	Stores Roll Number → Course mapping.
Iterator	Traverses collections and displays their contents.
Generics	DataManager<T> allows common operations for different object types.
Scanner	Accepts user input from the keyboard.
Menu Driven Program	Allows the user to select Add, Display, Search, Update, Delete and other operations.

3. JAVA PROGRAM

```
import java.util.*;

class Student {
    int roll;
    String name, dept, course; double cgpa;


    Student(int roll, String name, String dept, String course, double cgpa) {
        this.roll = roll; this.name =
        name; this.dept = dept;
        this.course = course; this.cgpa
        = cgpa;

    }

    public String toString() {
        return roll + " | " + name + " | " + dept + " | " + course + " | " + cgpa;
    }
}

class DataManager<T> {
    ArrayList<T> list = new ArrayList<>();

    voidadd(T x) {
        list.add(x);
    }

    voiddisplay() {
        Iterator<T> i = list.iterator(); while (i.hasNext())

            System.out.println(i.next());
    }

    voiddelete(int n) {
        if(n >= 0 && n < list.size())
            list.remove(n);
    }
}

public class StudentManagement {
      public static void main(String[] args) { Scanner sc = new
                                               Scanner(System.in);

        DataManager<Student> dm = new DataManager<>(); HashSet<String>
        departments = new HashSet<>(); HashMap<Integer, Double> marks = new
        HashMap<>(); Hashtable<Integer, String> courses = new Hashtable<>();


        while (true) {
            System.out.println("\n1.Add 2.Display 3.Search 4.Update"); System.out.println("5.Delete
            6.Departments 7.Roll-CGPA 8.Courses System.out.print("Enter choice: ");          9.Exit");

            int ch = sc.nextInt();
            sc.nextLine();

            if (ch == 1) {
                System.out.print("Roll No: "); int r =
                sc.nextInt(); sc.nextLine();


                System.out.print("Name: "); String n =
                sc.nextLine().trim();

                String[] words = n.toLowerCase().split(" "); String name = "";
```

```
        for (String w : words)
            if (!w.isEmpty())
                name += Character.toUpperCase(w.charAt(0))
                        +w.substring(1) + " ";
        name = name.trim();

        System.out.print("Department: ");
        String d = sc.nextLine();

        System.out.print("Course: ");
        String c = sc.nextLine();

        System.out.print("CGPA:  ");
        double g = sc.nextDouble();

        Student s = new Student(r, name, d, c, g);
        dm.add(s);
        departments.add(d);
        marks.put(r, g);
        courses.put(r, c);

        System.out.println("Words: " + name.split(" ").length);
        System.out.println("Characters: " + name.replace(" ", "").length());

        System.out.print("Enter substring to search: ");
        sc.nextLine();
        String sub = sc.nextLine();

        System.out.println(name.toLowerCase().contains(sub.toLowerCase())
                ? "Substring Found" : "Substring Not Found");
        System.out.println("Student added successfully.");

    } else if (ch == 2) {
        dm.display();

    } else if (ch == 3) {
        System.out.print("Enter Roll No: ");
        int r = sc.nextInt();
        boolean found = false;

        for (Student s : dm.list) {
            if(s.roll == r) {
                System.out.println(s);
                found = true;
            }
        }
        if(!found) System.out.println("Student Not Found");

    } else if (ch == 4) {
        System.out.print("Enter index to update: ");
        int i = sc.nextInt();
        sc.nextLine();

        if(i  >= 0 && i < dm.list.size()) {
            Student s = dm.list.get(i);

            System.out.print("New Name: ");
            s.name = sc.nextLine().trim();
            System.out.print("New Department: ");
            s.dept = sc.nextLine();
            System.out.print("New Course: ");
            s.course = sc.nextLine();
            System.out.print("New CGPA: ");
            s.cgpa = sc.nextDouble();
```

```
                System.out.println("Updated Successfully");
            }

        } else if (ch == 5) {
            System.out.print("Enter index to delete: ");
            dm.delete(sc.nextInt());
            System.out.println("Deleted Successfully");

        } elseif  (ch == 6) {
            Iterator<String> i = departments.iterator();
            while (i.hasNext())
                System.out.println(i.next());

        } elseif  (ch == 7) {
            Iterator<Map.Entry<Integer, Double>> i =
                    marks.entrySet().iterator();

            while (i.hasNext()) {
                Map.Entry<Integer, Double> e = i.next();
                System.out.println(e.getKey() + " -> " + e.getValue());
            }

        } elseif  (ch == 8) {
            Iterator<Map.Entry<Integer, String>> i =
                    courses.entrySet().iterator();

            while (i.hasNext()) {
                Map.Entry<Integer, String> e = i.next();
                System.out.println(e.getKey() + " -> " + e.getValue());
            }

        } else if (ch == 9) {
            System.out.println("Thank You!");
            break;
        }
    }
    sc.close();
  }
}
```

4. SAMPLE INPUT / OUTPUT

```
Enter choice: 1
Roll No: 101
Name:   aarathana priya
Department: IT
Course: Java
CGPA: 8.7
Enter substring to search: priya

Words: 2
Characters: 14
Substring Found
Student added successfully.
```

5. COLLECTION INTERFACES & JUSTIFICATION

List: Maintains an ordered collection and allows duplicate elements. ***ArrayList*** is selected because student records need indexed access and easy updates.

Set: Stores unique elements. ***HashSet*** is selected because department names must not be duplicated.

Map: Stores key-value pairs. ***HashMap*** is used for Roll Number–CGPA mapping, while ***Hashtable*** is used for Roll Number–Course details.

Generics: DataManager<T> provides type-safe common operations. ***Iterator:*** Traverses collections systematically. ***Enhanced For Loop:*** Simplifies traversal of the student list.

6. CONCLUSION

The application demonstrates String Handling and the Java Collection Framework in a menu-driven Student Management System. It uses List, Set and Map implementations, Generics, Iterator, Scanner and basic string-processing methods to store, process and display student information efficiently.